Homework — 1.4.3 Boolean Algebra

Name: ____ Date: _ ***Mark:*** ___ / 20

OCR H446 — set after the 1.4.3 lesson. SHOW ALL WORKING.

Part A — This week's topic (~14 marks)

1. Complete the truth table for the expression **Q = (A + B) · ¬C**. **[3]** *(AO2)*

A	B	C	A+B	¬C	Q
0	0	0			
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			

2. Use **De Morgan's law** and **double negation** to simplify `¬(A · ¬B)` so that no bar remains over more than one variable. Show each step and name the law used at each step. **[2]** *(AO2)*

Working space:

3. Simplify the expression `A + ¬A · B` as far as possible, naming the law(s) you use. **[2]** *(AO1/AO2)*

Working space:

4. The truth table below shows the output **X** of a logic circuit with inputs **A**, **B** and **C**.

A	B	C	X
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	1

(a) Complete the **Karnaugh map** for X. **[1]** *(AO2)*

	BC = 00	BC = 01	BC = 11	BC = 10
A = 0				
A = 1				

(b) Draw the groups on your Karnaugh map and use them to write the **simplified Boolean expression** for X. **[3]** *(AO2)*

Working space:

5. State the Boolean expressions for the **Sum** and **Carry** outputs of a **half adder**, and explain why a half adder **cannot** be used on its own to add the middle columns of a multi-bit binary addition. **[2]** *(AO1)*

Working space:

6. A **D-type flip-flop** is used in a circuit. State what a D-type flip-flop stores, and when its output is able to change. **[1]** *(AO1)*

Working space:

Part B — Retrieval: keep it fresh (~6 marks)

7. Explain the difference between **symmetric** and **asymmetric** encryption, and state **one** advantage of asymmetric encryption. *(1.3)* **[3]** *(AO1)*

Working space:

8. Describe how the **round robin** CPU scheduling algorithm works, and state **one** advantage of it. *(1.2)* **[3]** *(AO1)*

Working space: